Netcom Computers Pvt Ltd

Enterprise Employee Management Pro — Operations, Biometrics, Admin Punch, 1-Day Monthly Casual Leave & Deployment Manual

Company: Netcom Computers Pvt Ltd	**Database:** MySQL 8.0 + SQLite zero-friction fallback	**Biometrics:** Realtime TCP/UDP Port 4370 Sync
Host: 0.0.0.0:8000 (Local & LAN Access)	**Login Route:** /login/ (Strictly Preserved)	**Test Suite:** 32 Automated Tests (100% Pass)

1. System Dashboards, Duty Punch Terminals & Security Protections

• **Netcom Employee Dashboard:** Hero banner with profile picture avatar, login popup announcement with close button, photo + GPS attendance punch (strict once-per-day enforcement with button lock), personal calendar (strictly isolated to employee's own records), 4-record scroll containers.
• **Netcom HR Dashboard:** Leave review modal (Casual vs Sick Leave), daily work planning & staff notifications, comprehensive report downloads (PDF & Excel), HR self-service duty punch terminal, employee details with password toggle (■■), document links (■ View Doc), employee delete button, and 5-record scroll containers.
• **Netcom Admin Dashboard:** Administrator Duty Punch Terminal (webcam photo + GPS verification, real-time clock, instant punch in/out), full workforce governance, banner & popup image management, employee details with password reveal, submitted document inspection, biometric device sync, 10-record scroll containers.
• **Permanent Protection for Main Admin (NETCOM):** The main admin user NETCOM and superusers can NEVER be deleted. Delete buttons are completely hidden on frontend, and backend blocks deletion with an error.
• **Strict Access Enforcement:** HR users are completely blocked from accessing Admin pages (/dashboard/admin/) via backend @admin_required.

2. Leave Review, Office Paid Leave & Attendance Engine

Leave Action / Event	Review / Type	Balance Impact	Salary & Calendar Synchronization
Employee Applies for Leave	Triggers HR Notification	Balance held pending	Appears in HR Leave Requests; displayed on calendar with yellow pending border.
Approved: Paid Casual Leave	Paid Casual Leave	**-1 Casual Leave Day**	**Zero salary deduction. Displayed with GREEN border on calendar & dashboard.**
Approved: Paid Sick Leave	Paid Sick Leave	**Zero casual deduction**	**Zero salary deduction. Displayed with GREEN border on calendar & dashboard.**
Mark Paid Leave for Office	Office-Wide Holiday	**0 deduction from casual balance**	**Full pay (zero salary deduction). Calendar highlights date with purple badge & tooltips.**
Rejected Leave Request	Rejected with reason	Zero balance deduction	**Displayed in RED on employee calendar with clear rejection reason.**
Casual Leave Restoration	Auto-triggered on Punch/Biometric	**+1 Casual Leave Day restored**	**Leave cancelled, attendance marked Present/Half Day, audit log saved.**
Strict Once-Per-Day Punch	Backend & JS Button Lock	N/A	**Blocks duplicates with exact messages: *'You have already punched in/out for today.'***

3. Daily Work Planning, Profile Pictures, Passwords & HR Reporting

• **HR & Admin Password Inspection:** Authorized Admin and HR users can inspect employee login passwords via interactive ■■ Show/Hide toggle buttons across all employee management tables, edit modals, and profile dossiers.
• **Profile Pictures & Document Repository:** Profile picture uploaded during employee onboarding is rendered directly across dashboard hero headers, sidebar avatars, and directories. Submitted proof documents can be previewed or downloaded anytime by HR and Admin via ■ View Doc links.
• **Calendar Isolation:** Employees can strictly view only their own attendance calendar. HR and Admin can select any user (Employee, HR, Admin) to view their full month calendar.
• **Office Paid Leave Scheduling:** HR and Admin can select any date on their dashboard calendar to mark an official Paid Leave for the Office.
• **Employee Daily Work Planning:** HR users schedule next-day tasks, set dates, and send direct dashboard notifications to specific employees.
• **Comprehensive HR Report Export:** HR users select any date range and download full HR data (directory, attendance, leaves, balances, and daily work) in both **Excel (.xlsx)** and **PDF** formats.
• **Login Route Integrity:** Login URL /login/ is strictly preserved with clean layout and 'Employee Management Portal' title.

4. Authorized Access Credentials

Role	Username	Password	Code	Portal Features
Administrator	NETCOM	Netcom@2026@Netcom	ADM001	Netcom Admin Dashboard, HR Operations, Banner uploads, User details/leave balance search & edit, Biometric sync, Payroll
HR Manager	hr	hr12345	HR001	Netcom HR Dashboard, Leave review modal (Paid vs Unpaid), Work image previews, Duty punch, Biometric sync
Employee	emp	emp12345	EMP001	Netcom Employee Dashboard, Announcement popup, Webcam & GPS punch, Calendar with red rejected leaves, Work reports with image upload

5. Complete Project File Structure

```
employee_management/
■
■■■ manage.py
■■■ requirements.txt
■■■ .env
■■■ .gitignore
■
■■■ config/
■   ■■■ __init__.py
■   ■■■ settings.py
■   ■■■ urls.py
■   ■■■ asgi.py
■   ■■■ wsgi.py
■
■■■ apps/
■   ■■■ __init__.py
■   ■■■ employees/
■   ■   ■■■ models.py, views.py, urls.py, forms.py, admin.py
■   ■■■ departments/
■   ■■■ attendance/
■   ■   ■■■ models.py (DailyWork work_image & is_offsite, BiometricDevice)
■   ■   ■■■ services.py (Casual Leave restoration engine)
■   ■   ■■■ biometric_service.py (Hardware TCP/UDP socket sync)
■   ■   ■■■ views.py, urls.py, forms.py
■   ■■■ leave_management/
■   ■   ■■■ models.py (HRNotification, is_paid, approval_type)
■   ■   ■■■ views.py, urls.py, forms.py
■   ■■■ payroll/ (Full, Half Day, 2h Permission & Paid/Unpaid Leave calculation)
■   ■■■ users/ (decorators.py: @admin_required, @hr_or_admin_required)
■   ■■■ dashboard/ (Netcom Employee/HR/Admin Dashboards & Banner models)
■   ■■■ reports/
■
■■■ templates/
■   ■■■ base.html (Role-based navigation bar)
■   ■■■ employees/, departments/, attendance/, leave_management/, payroll/
■   ■■■ dashboard/ (employee_dashboard, hr_dashboard, admin_dashboard)
■
■■■ static/
■   ■■■ css/ (styles.css: Red rejected leave, scroll containers, lightboxes)
■   ■■■ js/ (app.js: Webcam, GPS, lightbox modals, leave review modals)
■   ■■■ images/
■
■■■ media/
■   ■■■ banners/ (Customizable dashboard hero banners)
■   ■■■ employee_photos/ & punch_photos/
■   ■■■ daily_work_images/ (Employee work activity attachments)
■
■■■ database/
■   ■■■ schema.sql
■   ■■■ backups/ (Full JSON database backups)
■
■■■ logs/
■■■ tests/
    ■■■ test_all_features.py (15 Automated Comprehensive Tests)
```

6. Display Limits & Scrolling Standards

Dashboard Section	Dashboard	Initial Records	Scrolling Behavior
My Daily Work Details	Employee	**4 records**	Vertical scroll enabled (max-height: 255px)
My Leave Details	Employee	**4 records**	Vertical scroll enabled (max-height: 255px)
Leave Requests	HR	**5 records**	Vertical scroll enabled (max-height: 310px)
Employee Work Details	HR	**5 records**	Vertical scroll enabled (max-height: 310px)
All Leave Requests	Admin	**10 records**	Vertical scroll enabled (max-height: 520px)
All Work Updates	Admin	**10 records**	Vertical scroll enabled (max-height: 520px)
Employee Details & Leave Balance	Admin	**10 records**	Vertical scroll enabled (max-height: 520px)
Employee & HR Attendance Details	Admin	**10 records**	Vertical scroll enabled (max-height: 520px)

7. Quick Start Execution & Railway Deployment Commands

```
# Local / LAN Execution:
python manage.py runserver 0.0.0.0:8000

# Execute 100% Automated Test Suite (33 comprehensive tests):
python manage.py test tests

# Railway Cloud Deployment:
# Configured via nixpacks.toml, Procfile, railway.json, Dockerfile, and WhiteNoise.
# Nixpacks automatically initializes Python 3.12, creates /opt/venv, and installs pip packages.
# Connect your Git repository to Railway, and Railway automatically migrates and serves via Gunicorn.
```